Eenheid 3.4

Die kettingreël

Stewart p 197

- $\frac{d}{dx}(6-5x)^2 = \frac{d}{dx}\left(36 - 60x + 25x^2\right)$ Plan!

$= -60 + (25)(2x) = -60 + 50x$

Maar??

- $\frac{d}{dx}(6-5x)^{100}$??

Saamgestelde funksie

Kettingreël – dié reël van alle differensiasiereëls
(James Gregory)

- $\frac{d}{dx} f(g(x))$ (saamgestelde funksies)

$= f'(g(x)) \times g'(x)$ *

Laat $(6-5x)^2 = (f \circ g)(x)$ (saamgestelde funksie)

$f(x) = x^2$

$f'(x) = 2x$

$g'(x) = -5$

- $g(x) = 6-5x$

$$\frac{d}{dx}(f\circ g)(x) = f'(g(x)) \times g'(x)$$

NB

$$= 2(g(x)) \times g'(x)$$

$$= 2(6-5x) \times -5 = -10(6-5x)$$

$$= -60 + 50x$$

Ander Notasie: Leibniz (substitusie)

Laat $y = f(g(x)) = f(u)$ waar $g(x) = u$

dan $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

Nou $(f\circ g)(x) = f(g(x)) = (6-5x)^2$

$= u^2$

STEL: $g(x) = (6-5x)$

Leibniz: $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$= 2u \times \frac{du}{dx} = 2(6-5x) \times \frac{d}{dx}(6-5x)$

$= 2(6-5x) \times -5$

$= -60 + 50x$

VOORBEELDE: A. afgeleides van trig. funksies

① $\frac{d}{dx} \cos x = -\sin x$

② $\frac{d}{dx} 2\cos x = 2\frac{d}{dx}\cos x = -2\sin x$

③ $\frac{d}{dx} \cos(2x) = -\sin(2x) \times 2$

④ $\frac{d}{dx} \cos(x^3) = -\sin(x^3) \times 3x^2$

⑤ $\frac{d}{dx} \cos^3 x = \frac{d}{dx} (\cos x)^3$

$= 3(\cos x)^2 \times -\sin x$

B: Vind die afgeleides van:

① $\frac{d}{dt} \frac{1}{\sqrt{t}}$

of $\frac{d}{dt} (t)^{-\frac{1}{2}} = -\frac{1}{2} t^{-\frac{1}{2}-1} = -\frac{1}{2} t^{-\frac{3}{2}}$

(2) $\frac{d}{dt}\frac{1}{\sqrt{1-t}} = \frac{d}{dt}(1-t)^{-1/2} = -\frac{1}{2}(1-t)^{-3/2} \times -1$

(3) $\frac{d}{dt}\frac{1}{\sqrt{\tan t}} = \frac{d}{dt}(\tan t)^{-1/2} = -\frac{1}{2}(\tan t)^{-3/2} \times \sec^2 t$

(4) $\frac{d}{dt}\tan\left(\frac{1}{\sqrt{t}}\right) = \sec^2\left(\frac{1}{\sqrt{t}}\right) \times -\frac{1}{2}t^{-3/2}$

C. Afgeleides van e^x

(1) $\frac{d}{dx}e^x = e^x$

② $\frac{d}{dx} e^{x^2} = e^{x^2} \times 2x$

③ $\frac{d}{dx} \sin(e^{2x}) = \cos(e^{2x}) \times e^{2x} \times 2$

④ $\frac{d}{dx} e^{\sin 3x} = e^{\sin(3x)} \times \cos 3x \times 3$

* Sien bewys $\frac{d}{dx}(e^{x}) = e^{x}$ einde v. lesingnotas

4 Stelling

Stewart p 202

- $\frac{d}{dx}\left(a^x\right) = a^x \ln a$ vir $a > 0$

Bv. $a = 2$

$\frac{d}{dx}\left(2^x\right) = 2^x \ln 2$

getal.

Bewys:

- Voorafkennis

"Hersiening": $e^{\ln a} = a$ (*) vir $a > 0$ (kanselleer eienskap)

(gebruik)

getal

$LK = \frac{d}{dx}\left(a^x\right) = \frac{d}{dx}\left(e^{\ln a}\right)^x = \frac{d}{dx} e^{(\ln a)(x)}$

$= e^{(\ln a)(x)} \times \frac{d}{dx}\left((\ln a)(x)\right)$

$= \left[e^{(\ln a)x}\right] \times \ln a = a^x \times \ln a = RK$

D Vind die afgeleides van:

① $\frac{d}{dt} 4^t = 4^t \ln 4$

② $\frac{d}{dx}\left(-4^{-x}\right) = -\frac{d}{dx}\left(4^{-x}\right)$

$= -4^{-x} \ln 4 \times \underbrace{\frac{d}{dx}(-x)}_{=-1}$

$= 4^{-x} \ln 4$

③ $\frac{d}{dx} 3^e = 0$

(Konstant)

$\frac{d}{dx}(a^x)$ (970)

$= a^x \ln a$

- $\frac{d}{dx}\left(a^{f(x)}\right)$

$= a^{f(x)} \ln a \times f'(x)$

④ $$\frac{d}{du}\operatorname{cosec}\left(5^{u}\right)$$ ① ②

$$= -\operatorname{cosec}\left(5^{u}\right)\cot\left(5^{u}\right) \times 5^{u}\ln 5$$

⑤ $$\frac{d}{dz}2^{\cot z} = 2^{\cot z}\ln 2 \times \frac{d}{dz}(\cot z)$$

$$= 2^{\cot z}\ln 2 \times -\operatorname{cosec}^{2} z$$

Kettingreël met som, produk, kwosiëntreël

Som

① $\frac{d}{dt}\left(3e^{t^2} - 6\cos\sqrt{t}\right) = 3e^{t^2} \times 2t - 6\left(-\sin\sqrt{t}\right) \times \frac{1}{2\sqrt{t}}$

(produkreël)

② $\frac{d}{dx}\left(\overset{f}{\sqrt{x^2-x}}\right) \times \left(\overset{g}{\sin x^2}\right) = f' \times g + f \times g'$

$= \left[\frac{1}{2}(x^2-x)^{-1/2}(2x-1)\right] \times \sin x^2 + \sqrt{x^2-x} \times \left[\cos x^2 \times 2x\right]$

③ $\frac{d}{dt}\left(\frac{4}{1+2t}\right) = \frac{d}{dt}\, 4 \times (1+2t)^{-1} = 4\frac{d}{dt}(1+2t)^{-1}$

$= (4)(-1)(1+2t)^{-2} \times 2$

? kwosiënt

skryf as produk

(4) $\dfrac{d}{dx}\left(\dfrac{e^{2x}}{x-\tan 3^{x}}\right)$ boonste / onderste (kwosiënt reël) $\left(\dfrac{f}{g}\right)' = ?$

$$= \frac{(e^{2x})(2)(x-\tan 3^{x}) - (e^{2x})(1-\sec^{2}(3^{x}) \cdot 3^{x}\ln 3)}{[x-\tan 3^{x}]^{2}}$$

(5) $\dfrac{d}{dx}\left(\dfrac{4}{e^{2x}+5^{3x}}\right)$ Skryf as produk

$$= \frac{d}{dx}\, 4 \times (e^{2x}+5^{3x})^{-1}$$

$$= (4)(-1)(e^{2x}+5^{3x})^{-1} \times (e^{2x} \times 2 + 5^{3x}\ln 5 \times 3)$$

(6) $\dfrac{d}{dx}(3x^{2}-7x)^{1/3}$

$$= \frac{1}{3}(3x^{2}-7x)^{-\frac{2}{3}}(6x-7)$$

Bepaal die afgeleide van die funksies en Moenie antwoorde (Oefen vbde) Vereenvoudig nie

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Notasie:

$$\frac{dy}{dx} = f'(x) = y'$$

① $y = e^{2x}$

$y' = e^{2x} \times 2$

② $f(x) = \sin e^{2x}$

$f'(x) = \cos e^{2x} \times e^{2x} \times 2$

③ $y = e^{\sin x^3}$

$y' = e^{\sin x^3} \times \cos x^3 \times 3x^2$

④ $f(t) = \frac{1}{t^4} = t^{-4}$

$f'(t) = -4t^{-5}$

⑤ $y = (t - t^3)^{10}$

$y' = 10(t - t^3)^9 \times (1 - 3t^2)$

⑥ $f(t) = \cos^4 3x = (\cos 3x)^4$

$f'(t) = 4(\cos 3x)^3 \times -\sin 3x \times 3$

⑦ $f(x) = 2^x$

$f'(x) = 2^x \ln 2$

(8) $f(x) = 2^{1-x}$

$f'(x) = 2^{1-x} \ln 2 \times (-1)$

(9) $f(x) = \cot(2^x)$

$f'(x) = -\operatorname{cosec}^2(2^x) \times 2^x \ln 2$

(10) $f(t) = 7\tan 3t - 3^{\tan t}$

$f'(t) = 7\sec^2 3t \times 3 - 3^{\tan t} \ln 3 \times \sec^2 t$

(11) $f(x) = \operatorname{cosec} \pi/3 = \frac{2}{\sqrt{3}}$ konstant

$f'(x) = 0$

12 $f(x) = \dfrac{\sqrt{x^3+1}}{7-\tan x}$ kwosiëntreël

$f'(x) =$

$$\frac{\frac{1}{2}(x^3+1)^{-1/2}(3x^2)(7-\tan x) - \left(\sqrt{x^3+1}\right)(-\sec^2 x)}{[7-\tan x]^2}$$

Te Bewys: $\frac{d}{dx}(e^x) = e^x$

Bewys:

Laat $f(x) = e^x$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{e^{x+h} - e^x}{h}$$

$$= \lim_{h \to 0} \frac{e^x . e^h - e^x}{h}$$

$$= \lim_{h \to 0} e^x \times \lim_{h \to 0} \frac{e^h - 1}{h}$$

$$= e^x \times \lim_{h \to 0} \frac{e^h - 1}{h}$$

Wat nou van $\lim_{h\to 0} \frac{e^h - 1}{h}$?

Nou $\underset{3}{f'(0)} = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$

$= \lim_{h\to 0} \frac{e^{0+h} - e^0}{h}$

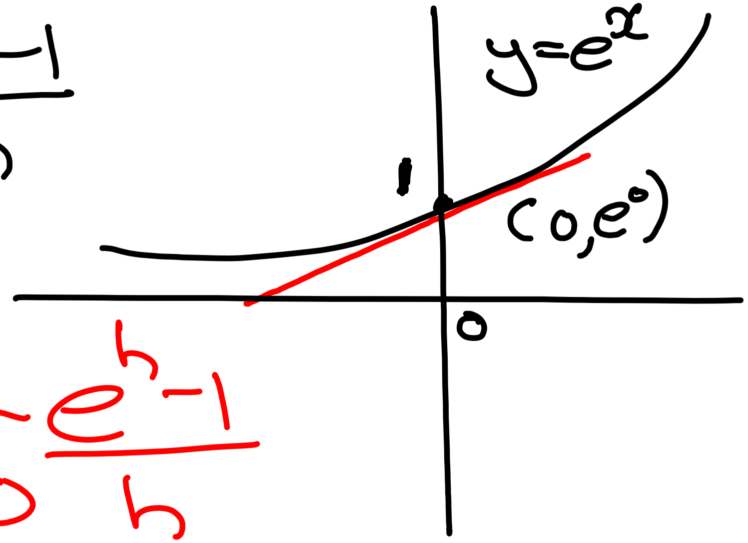
$= \lim_{h\to 0} \frac{e^h - 1}{h}$

Onthou: e is gedefinieer as die getal waar die helling v/d raaklyn by $x = 0$ gelyk aan 1 is.

* En helling van raaklyn is $f'(0) = 1$

Dan is $f'(0) = \lim_{h \to 0} \frac{e^h - 1}{h}$

$= 1$



Sodat $\frac{d}{dx}(e^x) = e^x \times \lim_{h \to 0} \frac{e^h - 1}{h}$

$= e^x \times 1$

$= e^x$